

Carboxylic acids and derivatives

A Level Chemistry

Carboxylic acids

A **carboxylic acid** 羧酸 has the -COOH (carboxyl) functional group. It is a weak acid.

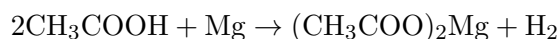
Making carboxylic acids

- **oxidation** 氧化 of a primary **alcohol** 醇 or an **aldehyde** 醛 with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or KMnO_4 , with **reflux** 回流 (so it is fully oxidised).
- **hydrolysis** 水解 of a **nitrile** 腈 with dilute acid or alkali, then acidifying.
- hydrolysis of an **ester** 酯 with dilute acid or alkali and heat, then acidifying.

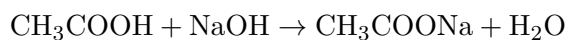
Reactions of carboxylic acids

These reactions all show that carboxylic acids are acids:

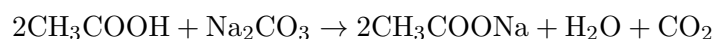
- **with a reactive metal** (a **redox** 氧化还原 reaction): gives a **salt** 盐 and hydrogen.



- **with an alkali** (a **neutralisation** 中和): gives a salt and water.



- **with a carbonate** 碳酸盐: gives a salt, water and carbon dioxide. The fizzing of CO_2 is a test for a carboxylic acid.



Two reactions change the functional group:

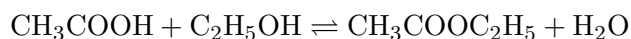
- **esterification** 酯化 with an alcohol (concentrated H_2SO_4 catalyst) gives an ester.
- **reduction** 还原 by LiAlH_4 gives a primary alcohol.

Esters

An ester has the -COO- group. It often smells sweet or fruity.

Making esters

An ester forms in a **condensation** 缩合 reaction between an alcohol and a carboxylic acid, with concentrated H_2SO_4 as catalyst. A water molecule is lost, and the reaction is reversible:



Hydrolysis of esters

Hydrolysis splits the ester back apart. The conditions change the products:

- **dilute acid** and heat: reversible. Gives back the carboxylic acid and the alcohol.
- **dilute alkali** and heat: not reversible. Gives the alcohol and the **salt** of the carboxylic acid (the carboxylate ion).